

KARNATAKA RESIDENTIAL EDUCATIONAL INSTITUTIONS SOCIETY

ICT Class 9 - Formative Assessment 2

Formative Assessment (FA-2)

Class: Class 9

Assessment Type: Formative Assessment (FA-2)

Total Marks: 30

Duration: 90 minutes

Topics Covered: Data Structures - Lists, Tuples, Dictionaries, Sets

General Instructions

1. All questions are compulsory.
2. Read each question carefully before answering.
3. Write neatly and legibly.
4. Marks are indicated against each question.

Part A: Written Concept Paper (20 Marks)

Section I: MCQ (5 x 1 = 5 marks)

a. What is the output?

```
s = {1, 2, 3, 2, 1}
print(len(s))
```

- a. 5
- b. 3
- c. 2
- d. Error

b. Which method returns a list of tuples representing key-value pairs in a dictionary?

- c. keys()
- d. values()
- e. items()
- f. pairs()

c. What will be the output?

```
lst = [[1, 2], [3, 4], [5, 6]]
result = [x for row in lst for x in row if x % 2 == 0]
print(result)
```

- a. [1, 3, 5]
- b. [2, 4, 6]
- c. [2, 4]
- d. [[2], [4], [6]]

d. Which of the following is true about tuples?

- e. Tuples are mutable
- f. Tuples can contain mixed data types
- g. Tuples are faster than lists
- h. Both B and C

e. What does `dict.fromkeys(['a', 'b', 'c'], 0)` return?

f. {'a': 0, 'b': 0, 'c': 0}

g. {'a': 'b': 'c': 0}

h. ['a', 'b', 'c']

i. Error

Section II: Short Answer (5 x 2 = 10 marks)

a. Differentiate between a **list** and a **tuple**. Give two real-world use cases where a tuple is preferred over a list.

b. Write the output of:

```
d1 = {"x": 1, "y": 2}
d2 = {"y": 3, "z": 4}
d1.update(d2)
print(d1)
```

c. Explain the difference between `remove()`, `pop()`, and `del` for lists with examples.

d. Write a Python program to find the intersection and union of two sets without using built-in set operations.

e. What is **list comprehension**? Write a list comprehension to generate all Pythagorean triplets (a, b, c) where $a < b < c$ and $a^2 + b^2 = c^2$ for values up to 20.

Section III: Long Answer (5 x 1 = 5 marks)

a. Write a Python program that implements a **word frequency counter** using dictionaries. The program should:

- Read a paragraph of text
- Count the frequency of each word (case-insensitive, ignoring punctuation)
- Sort words by frequency in descending order
- Display the top 5 most frequent words with their counts
- Find and display words that appear exactly once

Part B: Hands-on Practical Lab Task (5 marks)

Write a Python program that:

- Creates a list of 15 student records as dictionaries with keys: name, roll_no, marks (out of 100)
- Implements functions to:
 - Find the topper (highest marks)
 - Calculate the class average
 - Find students who scored below average
 - Sort students by marks in descending order
 - Display results in a formatted table

Part C: Notebook Maintenance (5 marks)

- a. Neatness and organisation of notebook (2 marks)
- b. Completion of all assigned classwork and homework (2 marks)
- c. Proper headings, dates, and indexing (1 mark)